Exploring the Beauty Industry: Cosmetic Networks

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01

Intro/Motivation

Beauty Industry

- Global Cosmetics Market is worth around $350 billion
- People wear makeup everyday!
- Different skin types, colors
- Which chemicals are being applied to your face?



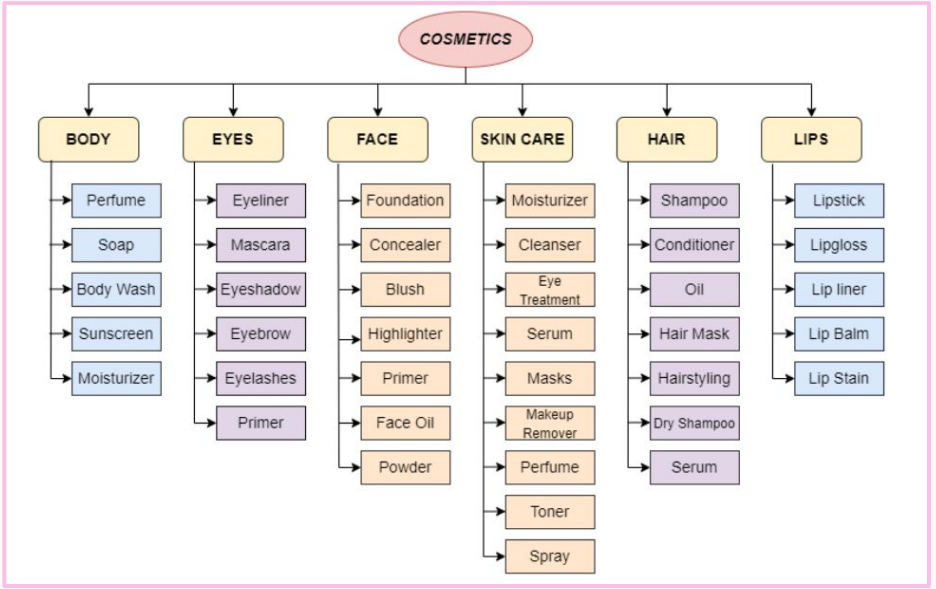
02

Data

Dataset

E-Commerce Cosmetic Products (Kaggle, 2024)

- 11000+ cosmetic products scraped from Amazon, Sephora, Ulta +
- Product, Ingredients, Category, Cost, Ranking
- Filter to Eyes, Face, & Lips



03

Exploration

Research Questions

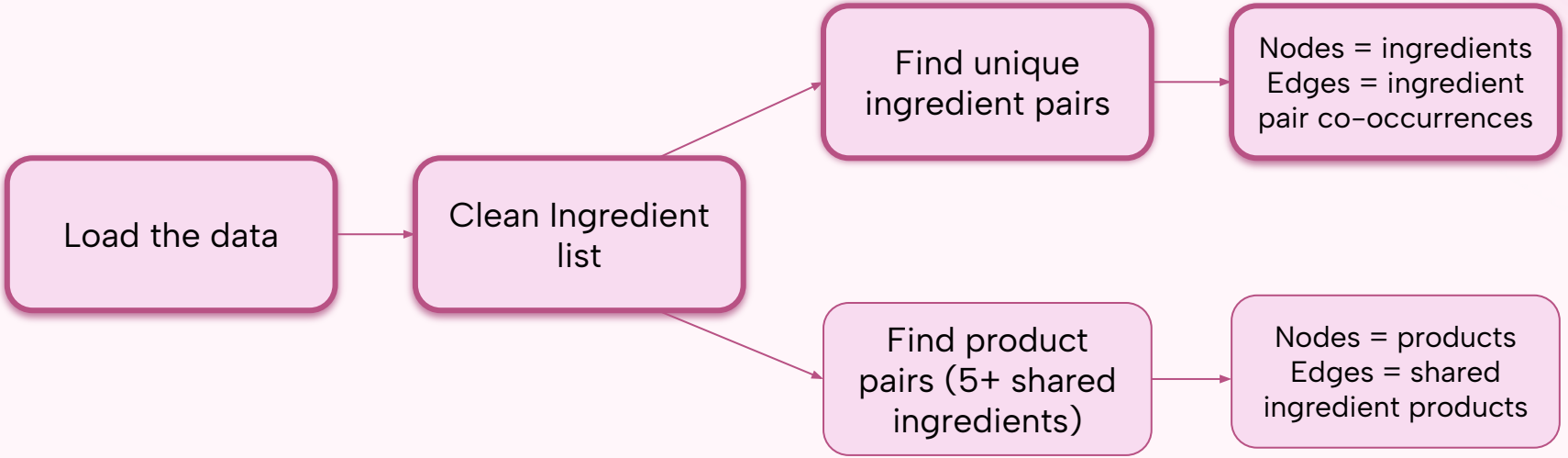
- What are the most common ingredients across cosmetics?
- Which products are most similar in formulation?
- What categories of cosmetics have the most shared ingredients?
- Are there alot of toxic chemicals in cosmetics?



04

Networks

Network Creation Pipeline



Ingredient Co-Occurrence Network

Complete Network

- 4119 Nodes (Chemical Ingredients)
- 183355 Edges (Co-Occurrences)
- 26 Communities

Filtered Network by Avg. Weighted Degree & Backbone Edges

- 1383 Nodes (33.576% Visible)
- 35107 (31.3% Visible)
- 21 Communities

Ingredient Co-Occurrence Network

Nodes: Chemicals
Edges: Co-occurrence in products

Num. Nodes: 1383
Edges: 35017 (31.3% Visible)

Orange Community: Pigments & Mineral Fillers

Pink Community: Core Formulation

Brown Community: Antioxidants & Fragrance



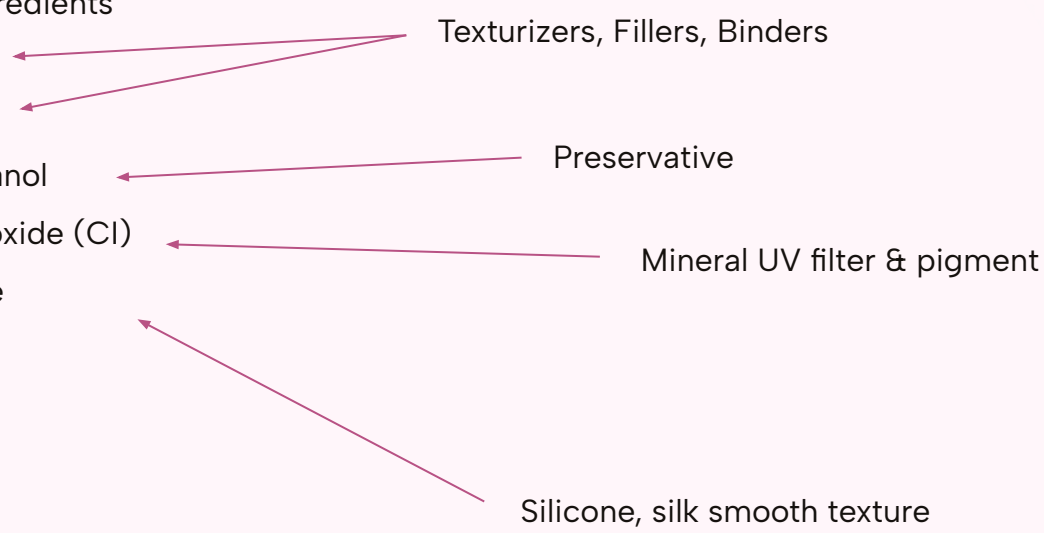
Ingredient Co-Occurrence Network

Rank	Weighted Degree	Betweenness	PageRank	Eigenvector
1	silica	phenoxyethanol	phenoxyethanol	phenoxyethanol
2	mica	silica	silica	silica
3	phenoxyethanol	tocopherol	mica	caprylyl glycol
4	titanium dioxide (ci)	mica	titanium dioxide (ci)	tocopherol
5	dimethicone	dimethicone	tocopherol	tocopheryl acetate
6	tocopherol	tocopheryl acetate	caprylyl glycol	mica
7	caprylyl glycol	retinyl palmitate	dimethicone	dimethicone
8	tocopheryl acetate	caprylyl glycol	ci	butylene glycol

Ingredient Analysis

Most Common Ingredients

- Silica
- Mica
- Phenoxyethanol
- Titanium Dioxide (CI)
- Dimethicone



Toxic Chemicals

Community 14 Identified the following:

Globally Banned:

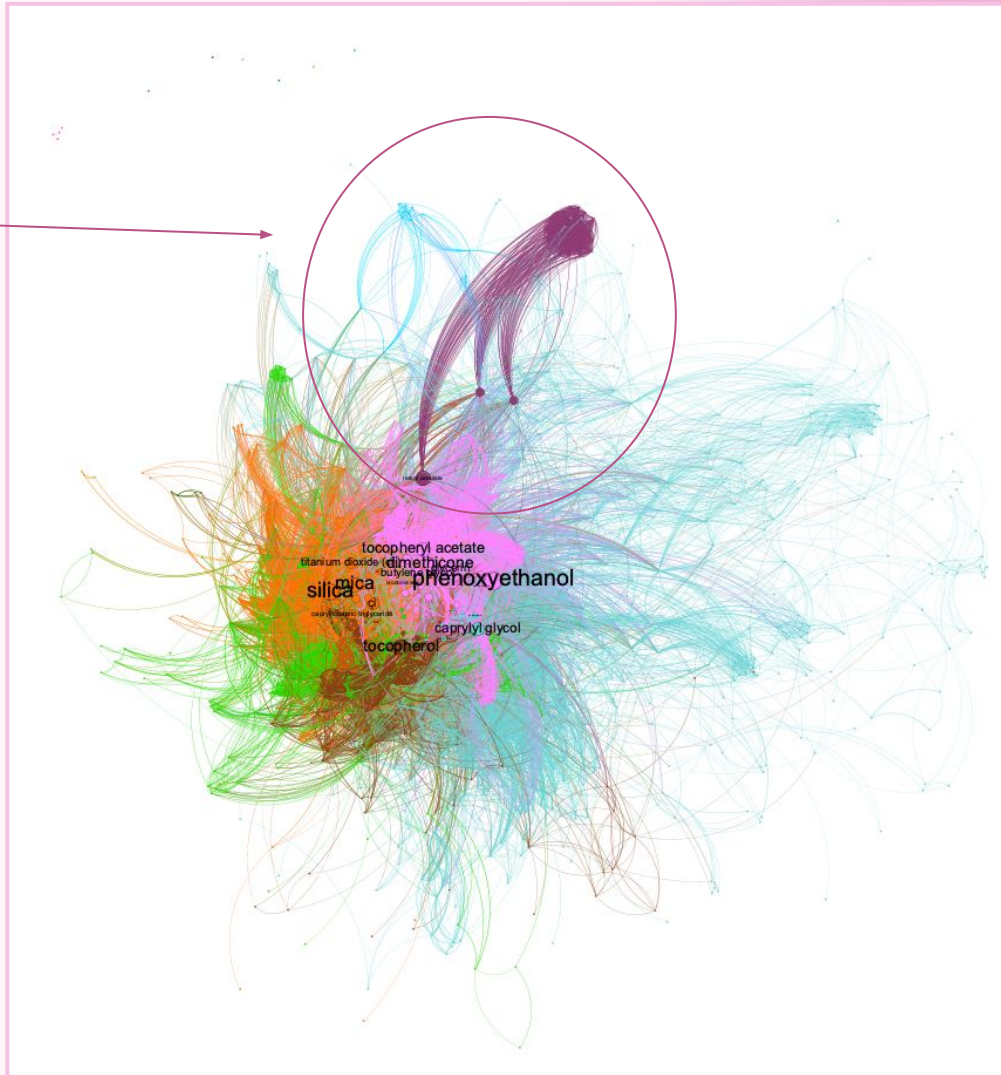
- Lead
- Mercury

Restricted:

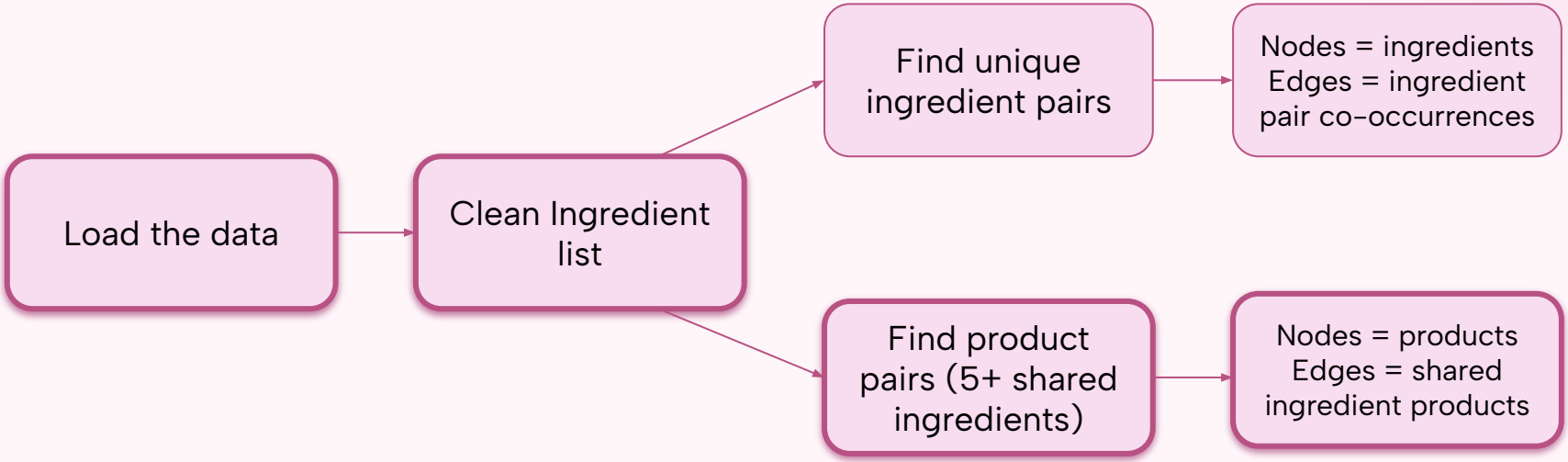
- Parabens
- Oxybenzone

Clean Beauty Avoids:

- Sulfates
- Talc



Network Creation Pipeline



Product-Product Network

Nodes: Products
Edges: Product share >13 ingredients

Num. of Nodes: 2787
Edges: 32080 (4.13% Visible)
Modularity: 0.753

Dark Pink Community: Lipstick/Lipgloss

Light Pink Community: Concealers

Green Community:
Concealers/Blush/Eye Primer/Lipstick

Blue Community:
Eyeshadows/Eyeliner/Lips



Product-Product Network

Rank	Weighted Degree	Betweenness	PageRank	Eigenvector
1	**CLINIQUE** All-Over Concealer + Corrector	**Dior** Eyeliner	**Smashbox** Blush	**CLINIQUE** All-Over Concealer + Corrector
2	**CLINIQUE** Blush	**FENTY** Semi-Matte Lipstick	**CLINIQUE** Lipstick - Blush	**CLINIQUE** Blush
3	**CLINIQUE** Blush	**Smashbox** Blush	**CLINIQUE** Lipstick - Nuzzle	**CLINIQUE** Blush
4	**CLINIQUE** Blush	**MAC:**Pro Longwear Eyeliner	**CLINIQUE** Lipstick - Subtle	**CLINIQUE** Blush
5	**CLINIQUE** Blush	**MAC:**Locked Kiss Ink Lipstick	**CLINIQUE** Lipstick	**CLINIQUE** Blush

What is Clinique?

- Dermatologist Guided – Safe Formulas
- Clean Ingredients
- Simple & 100% Fragrance-Free



Lipstick Ingredients
Moisturizing Agents: Ricinus Communis (Castor) Seed Oil, Butyrospermum Parkii (Shea Butter).

Conditioning/Nourishing: Tocopheryl Acetate (Vitamin E), Centella Asiatica (Hydrocotyl) Extract, Hibiscus Abelmoschus Extract.

Structure & Texture: Ozokerite, Polyethylene, Microcrystalline Wax, Silica.

Colour & Finish: Palmitoyl Tripeptide-1 (for structure) and a complex of waxes.

Key Takeaways & Limitations

- Most makeup share a chemical backbone of preservatives, emulsifiers, and stabilizers
- Lip & Eye products have the most similar ingredients across categories
- Clean makeup brands apply a consistent chemistry across product categories
- Heavy metals and restricted compounds are more prevalent in metallic & shimmery pigments (eyeshadows and lipsticks)

Thanks!

Questions?